

A Morphological and Computational Analysis of Yoruba Numeral Multiplicands: Linguistic Structure and Cultural Significance.

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Abstract

The Yoruba numeral system is a complex linguistic framework that extends beyond basic enumeration, integrating morphological structures with cultural and cognitive significance. This study presents a morphological and computational analysis of Yoruba numeral multiplicands, examining their systematic formation through compounding, multiplication, and recursive linguistic processes. Anchored in Item-and-Process (IP) Theory, it explores how these numerals evolve through structured morphological mechanisms rather than simple morpheme concatenation. Employing a qualitative methodology, the research draws on secondary data, including linguistic analyses, scholarly literature, and expert consultations. Yoruba multiplicands are categorized into basic and derived types, with particular attention to their structural patterns, derivational strategies, and computational modeling potential. The findings reveal a recursive, rule-governed system that supports the representation of numerical values from single digits to figures in the billions. Key morphological strategies include multiplicative compounding, subtractive derivation, and symbolic markers such as “GBẸ + RÚN” to denote exponential increments. This study contributes to linguistic typology by demonstrating how indigenous numeral systems maintain internal coherence while accommodating complex numerical expressions. It also highlights the influence of external linguistic contact and underscores the importance of preserving the integrity of Yoruba numerals in modern discourse. Ultimately, it offers insights into the intersection of morphology, numeration, and computation, affirming the Yoruba numeral system as a linguistically robust and culturally embedded framework.

Keywords: Item-and-Process Theory, Linguistic typology, Morphological analysis, Numeral multiplicands, Yorùbá numeral system

Introduction

The Yoruba numeral system exemplifies a sophisticated morphological structure that intertwines linguistic principles with cultural significance. As a foundational element of Yoruba language and tradition, numeration in Yoruba is not merely a method of counting but an intricate system shaped by deep-seated linguistic rules and cultural practices. This paper examines the morphology of Yoruba numeral multiplicands, categorizing them into basic and derived types while applying morphological theory to

analyze their formation. Yoruba numerals employ fundamental linguistic processes such as compounding, addition, and multiplication to represent values ranging from single digits to complex quantities in the millions. By investigating Yoruba multiplicands through a morphological lens, this study aims to illuminate the systematic principles that enable the Yoruba numerical system to function with both coherence and expansiveness.

Yoruba numeration has been a subject of scholarly inquiry due to its reliance on additive and multiplicative processes, which distinguish it from many Indo-European numeral systems. Foundational research by Gaye and Beecroft (1923), Bamgbose (1966), Ekundayo (1977), and Ajiboye (2013) highlights the structure and logic underlying Yoruba numerical expressions. While existing studies primarily classify Yoruba numerals and describe their syntactic behavior, this research extends the discussion by analyzing numeral multiplicands within the framework of morphological theory. By doing so, it provides deeper insights into the structural mechanisms that govern Yoruba numeration and its broader implications for linguistic typology and cognitive processing.

This study situates Yoruba numeral morphology within the context of linguistic complexity, exploring how morphological rules shape the language's approach to expressing numerical values. Understanding these mechanisms not only contributes to the documentation and analysis of an indigenous African language but also offers comparative insights into numeral systems worldwide. In doing so, this paper aims to bridge linguistic analysis with cultural understanding, demonstrating how Yoruba numerals reflect a unique interplay between mathematical concepts and linguistic evolution.

Literature Review

Several researchers have examined the Yoruba numeral system, contributing valuable insights into its structure, classifications, and sociolinguistic implications.

- **Gaye and Beecroft (1923)** provided one of the earliest classifications of Yoruba numerals, categorizing them into cardinal, ordinal, distributive, and adverbial forms. They described how each type functions within the language but did not extensively analyze the construction of derived forms.
- **Bamgbose (1966)** expanded upon this classification by exploring Yoruba numerals' role within nominal groups, describing their morphological features and syntactic arrangements. He identified that

numerals could extend beyond mere counting and be used structurally within sentences. However, his analysis focused more on surface forms and lacked depth in exploring numeral derivations.

- **Ekundayo (1977)** introduced the concept of complex numerals within Yoruba, describing how Yoruba numerals exhibit creativity and variability through operations such as addition, subtraction, and multiplication. He stated how a particular number could be expressed in different forms by giving example of number 19,669 that was expressed in seven different forms. He classified Yoruba numerals into basic and complex forms and discussed how the derived forms allow for intricate expressions. Ekundayo highlighted Yoruba's unique ability to represent complex numbers through its multiplicative structure, though he did not delve into the morphological analysis of each multiplicand.
- **Ajiboye (2013)** provided a morphological analysis of Yoruba numerals, dividing them into basic and derived forms and categorizing derivations into additive, subtractive, and multiplicative processes. Ajiboye's work emphasized the combination of basic numerals to generate complex ones, yet his study did not provide extensive examples of higher-order multiplicands.
- **Olubode-Sawe (2013)** supported the division of Yoruba numerals into basic and derived forms, emphasizing the importance of addition, subtraction, and multiplication in their formation. She acknowledged the influence of phonological patterns in the formation of complex numerals but did not examine the multiplicands systematically, which this paper seeks to address.
- **Longe (2016):** Building on his earlier work from 2009, Longe (2016) introduced an innovative Yorùbá decimal system that utilizes Òkẹ́ (meaning "twenty-thousand") as a fundamental multiplicand for constructing large numbers up to one billion. This system simplifies numerical representation by breaking down large values using Òkẹ́, as demonstrated in calculations for one million (Áádóta Òkẹ́), thirty million (Èḗdégbèjò Òkẹ́), and one billion (Èjì Òkẹ́-òkẹ́-lé-egbàarùn Òkẹ́). While linguistically permissible and aligned with Yorùbá numerical structures, Longe's approach posed challenges in expressing numbers beyond forty million efficiently. The multiplicand Òkẹ́ proved innovative but lacked scalability for more complex numerals, prompting the need for refinements. This study builds on Longe's contributions by enhancing the system for greater efficiency and reducing complexity in

large numeral formulations.

- **Ajiki (2022):** Ajiki (2022) classified Yorùbá numerals into two primary categories: **cardinal** and **ordinal** numbers. She further divided cardinal numbers into **integral** (basic) and **derived** numbers, where integral numbers serve as the foundational elements for forming complex numerals. Her classification emphasizes the use of specific multiplicands—ogún (20), igba (200), ẹgbàá (2,000), and Ọkẹ (20,000)—for constructing numbers up to one billion. Additionally, Ajiki introduced the concept of **doublet representation**, allowing numerals like 190 to be expressed in two ways (ẹwádínlúgba or igbádínmẹwàá), showcasing the system's flexibility. While her approach provides a structured framework, the doublet system may introduce unnecessary complexity. This study adopts Ajiki's classification but simplifies numeral representations to enhance usability and maintain precision.

Through these studies, it is evident that Yoruba numerals are both culturally significant and linguistically complex. However, previous research has primarily focused on general classifications rather than the specific morphological processes within multiplicands. This study builds upon these contributions by offering a detailed morphological analysis of Yoruba multiplicands, exploring how they support the language's capacity for expressing extensive numerical values.

3. Statement of the Problem

The Yoruba numeral system embodies remarkable linguistic complexity and cultural significance, yet significant gaps remain in understanding its morphological processes, particularly in relation to multiplicands. While previous research has classified Yoruba numerals into basic and derived forms, focusing on their use in compounding and additive structures, there has been limited exploration of the systematic structural and functional mechanisms underlying multiplicands.

A key challenge lies in the representation of extensive values, such as millions and billions, through derived multiplicands. While scholars like Longe (2009, 2016) and Ajiki (2022) adopted Ọkẹ (20,000) as a multiplicand for constructing numerals up to one billion, their approaches remain complicated and structurally inefficient beyond the million range. The absence of a simplified and systematic framework for large-number formation restricts a comprehensive understanding of how Yoruba numeration maintains both adaptability and coherence within linguistic

and cultural contexts.

Additionally, the increasing influence of external elements, such as loanwords and Western numerical representations, poses a challenge to the linguistic integrity of the Yoruba numeral system. Without a structured morphological analysis of how Yoruba numerals evolve and integrate external influences, the system risks losing its original coherence and adaptability in modern usage.

This study seeks to bridge these gaps by applying Item-and-Process (IP) Theory to Yoruba numeral multiplicands. Unlike previous descriptive classifications, this approach conceptualizes numeral formation as a series of systematic morphological processes rather than a mere linear combination of discrete morphemes. By doing so, this research aims to contribute both to morphological theory and to the preservation and functional optimization of the Yoruba numeral system, reinforcing its role as a linguistically and culturally resilient numerical framework.

4. Theoretical Framework

Item-and-Process (IP) Theory in Yoruba Numeral Multiplicands

Morphology, as a branch of linguistics, explores the internal structure of words and the rules governing their formation. The Item-and-Process (IP) Theory, one of the key morphological models, provides an effective analytical framework for understanding how Yoruba numeral multiplicands are formed and systematically expanded to generate complex numerical expressions. The IP approach, as proposed by Hockett (1954) and further developed by Anderson (1992), views word formation as a series of morphological processes applied to a base form rather than merely combining discrete morphemes in a linear arrangement. This theory is particularly useful in analyzing Yoruba numeral multiplicands, which rely heavily on compounding, addition, and multiplication to construct large numerical values.

Justification for the Application of IP Theory

Yoruba numerals exhibit morphological complexity that extends beyond simple root-based formations. Their structure is process-driven, involving systematic transformations rather than mere concatenation of morphemes. The IP theory is ideal for analyzing Yoruba numeral multiplicands because:

1. **It Accounts for Morphophonemic Changes:** Yoruba numerals undergo significant transformations when forming complex numbers.

For instance, the numeral for one million (Àádóta Òkẹ̀) is derived through a multiplicative process ($50 \times 20,000$). The IP model explains such derivations as successive applications of a base form (Òkẹ̀ = 20,000).

2. **It Explains Rule-Governed Operations:** Yoruba numerals utilize regular mathematical operations such as addition (lé “plus”), subtraction (dín “minus”), and multiplication (lónà “times”). These are processes rather than static morphemes, making IP theory a suitable model to explain their formation.
3. **It Accommodates Recursive Word Formation:** Yoruba numerals can be extended indefinitely using consistent derivational rules, as seen in structures like:
 - Èdédégbèjọ Òkẹ̀ $\rightarrow 30,000,000$ ($1,500 \times 20,000$)
 - Ègbègbèrún $\rightarrow 1,000,000$ (lit. “group of thousands”)

This recursive pattern aligns with the process-driven approach advocated by the IP theory.

The Item-and-Process (IP) Theory provides the most comprehensive framework for analyzing Yoruba numeral multiplicands. It effectively captures the **rule-governed morphological operations, recursive formations, and mathematical derivations** present in the Yoruba numeral system. By applying this theory, this study enhances our understanding of how Yoruba numerals evolve and how their structure remains **linguistically and culturally coherent** while supporting **large-scale numeral formation**.

5. Methodology

This study uses a qualitative, morphology-focused approach, analyzing Yoruba numeral multiplicands through the principles of compounding and inflectional morphology. Secondary sources, including academic publications and expert consultations, support the analysis. By categorizing Yoruba multiplicands into basic and derived types, the study provides a structured examination of the morphological mechanisms that facilitate large-scale numeral formation.

6. Yoruba Numeral Multiplicands: Structure and Examples

The Yoruba numeral system is structured around multiplicands, which serve as the building blocks for forming both basic and complex numbers. These multiplicands function through a combination of addition, subtraction, and multiplication, allowing Yoruba speakers to express a vast range of numerical

values efficiently. The numeral formation process adheres to systematic morphological principles, where base units interact with numerical markers to create higher-order values.

This section categorizes Yoruba numeral multiplicands into basic and derived forms. Basic multiplicands are foundational numbers that remain indivisible within the language's morphology, forming essential reference points for constructing larger numbers. In contrast, derived multiplicands result from systematic compounding and multiplication processes, allowing for the expansion of Yoruba numerals into the millions and beyond.

By analyzing these multiplicands, we gain deeper insights into how Yoruba numeration balances linguistic simplicity with mathematical complexity, ensuring clarity and consistency in large-number expressions.

6.1 Basic Multiplicands

Basic multiplicands are foundational morphemes representing fixed quantities. They are indivisible within Yoruba morphology, forming essential units for further derivation through compounding and multiplication.

- **Ogún** ('Twenty' or 20): Serves as a base for numbers from 40 to 200. Examples include *ogójì* (40) and *ogórùn-ún* (100).
- **Igba** ('Two Hundred' or 200): Extends from 400 to 2,000. Examples include *irinwó* (400) and *egbèrún* (1000)
- **Òkẹ** ('Twenty Thousand' or 20,000): Expands values from 20,000 to 200,000, such as *òkẹjì* ($20,000 \times 2 = 40,000$) and *òkẹrùn-ún* ($20,000 \times 5 = 100,000$).

These basic multiplicands operate as stand-alone morphemes, providing structural integrity to derived forms.

6.2 Derived Multiplicands

Derived multiplicands represent complex values formed by applying Yoruba's principles of compounding and multiplication. Using compounding, Yoruba numerals expand into larger values by joining base morphemes with multiplicative components, creating a wide numerical range.

1. Egbàá (Two Thousand)

- Formed by multiplying *igba* (200) by 10, resulting in *egbàá* (2,000). Compounding further allows variations up to 20,000.
- Examples:
 - **Èédégbàáta (5,000)** – six thousand minus one thousand.
 - **Egbàáta** (6,000) – two thousand multiplied by three
 - **Egbàárùn-ún** (10,000) – two thousand multiplied by five

2. **Ẹgbẹ̀gba (Two Hundred Thousand)**

- Derived from *ẹgbẹ̀rún lónà igba* (1,000 x 200). As a compound form, *ẹgbẹ̀gba* functions up to two million.
- Examples:
 - *Ẹ̀dẹ̀gbẹ̀gbẹ̀ta* (500,000) – six-hundred thousand minus one-hundred thousand.
 - *Ẹgbẹ̀gbẹ̀ta* (600,000) – two-hundred thousand multiplied by three.
 - *Ẹgbẹ̀gbẹ̀rún* (1,000,000) – two-hundred thousand multiplied by five.

3. **Ẹgbẹ̀gbàá (Two Million)**

- Created through compounding *ẹgbẹ̀gba lónà mẹwàá* (200,000 x 10), yielding values up to 20,000,000.
- Examples:
 - *Ẹ̀dẹ̀gbẹ̀gbẹ̀àtá* (5,000,000) – six million minus one million
 - *Ẹgbẹ̀gbẹ̀àtá* (6,000,000) – two million multiplied by three.
 - *Ẹgbẹ̀gbẹ̀àrún-ún* (10,000,000) – two million multiplied by five.

4. **Ẹgbòkẹ (Twenty Million)**

- Formed by multiplying *ẹgbẹ̀rún lónà ọkẹ* (1,000 x 20,000), allowing expressions from 20,000,000 to 200,000,000.
- Examples:
 - *Ẹ̀dẹ̀gbòkẹ̀ta* (50,000,000) – sixty million minus ten million.
 - *Ẹgbòkẹ̀ta* (60,000,000) – twenty million multiplied by three.
 - *Ẹgbòkẹ̀rún-ún* (100,000,000) – twenty million multiplied by five.

5. **Ẹgbẹ̀gbẹ̀bà (Two Hundred Million)**

- Created from *ẹgbẹ̀gbẹ̀rún lónà igba* (1,000,000 x 200), expressing values from 200,000,000 up to 2,000,000,000.
- Examples:
 - *Ẹ̀dẹ̀gbẹ̀gbẹ̀bẹ̀ta* (500,000,000) – six-hundred million minus hundred million.
 - *Ẹgbẹ̀gbẹ̀bẹ̀ta* (600,000,000) – two-hundred million multiplied by three
 - *Ẹgbẹ̀gbẹ̀bẹ̀rún* (1,000,000,000) – two-hundred million multiplied by five

6. **Ẹgbẹ̀gbẹ̀bàá ('Two Billion' or 2,000,000,000)**

- Derived from *Ẹgbẹ̀gbẹ̀bà lónà mẹwàá* (200,000,000 x 10), allowing expressions from 2,000,000,000 to twenty billion (20,000,000,000). Its odd version, *Ẹ̀dẹ̀gbẹ̀gbẹ̀bàá*, operates within three billion (3,000,000,000) to nineteen billion (19,000,000,000).
- Examples:

- 5,000,000,000 - Èédégbègbàáta (six billion minus one billion)
- 6,000,000,000 - Egbègbègbàáta (two billion multiplied by three)
- 10,000,000,000 - Egbègbègbàárùn-ún (two billion multiplied by five)
- 16,000,000,000 - Egbègbègbàájò (two billion multiplied by eight)

Following this multiplicand structure, further multiplicands like twenty billion (20,000,000,000) could be expressed as **Egbègbòké**, based on **Egbègbèrún lónà òké** (1,000,000 x 20,000), extending as far as two hundred billion (200,000,000,000) with **Egbègbègbègbà**; based on Egbègbègbèrún lónà igba (1,000,000,000 x 200) will run beyond one-trillion. As in

- 1,000,000,000,000 - Egbègbègbègbèrún 'One-Trillion'

7. Morphological Analysis of Yoruba Numeral Multiplicands Using Item-and-Process (IP) Theory

The Item-and-Process (IP) Theory provides a systematic approach to understanding how Yoruba numerals expand through multiplicative and subtractive processes. Yoruba numerals are not formed by merely arranging morphemes in a linear fashion but by applying systematic rules to base forms. The following analysis examines how Yoruba multiplicands are structured using the IP theory, which views numeral formation as a series of morphological transformations applied to base forms.

7.1 Formation Through Multiplication

Multiplication is a core process in Yoruba numeral morphology, where a base numeral undergoes a systematic transformation to form larger values. The multiplicative transformation follows a clear pattern:

- **Egbàá (Two Thousand) → Igba (200) × 10**
 - Egbàáta (6,000) → Egbàá × 3 (2,000 × 3)
 - Egbàárùn-ún (10,000) → Egbàá × 5 (2,000 × 5)
- **Egbègba (Two Hundred Thousand) → Egberun (1,000) × Igba (200)**
 - Egbègbeta (600,000) → Egbègba × 3 (200,000 × 3)
 - Egbègbèrún (1,000,000) → Egbègba × 5 (200,000 × 5)
- **Egbègbàá (Two Million) → Egbègba × 10**
 - Egbègbàáta (6,000,000) → Egbègbàá × 3 (2,000,000 × 3)
 - Egbègbàárùn-ún (10,000,000) → Egbègbàá × 5 (2,000,000 × 5)
- **Egbòké (Twenty Million) → Egberun (1,000) × Òké (20,000)**
 - Egbòkéta (60,000,000) → Egbòké × 3 (20,000,000 × 3)

- $\text{Ègbòkérùn-ún (100,000,000)} \rightarrow \text{Ègbòké} \times 5 (20,000,000 \times 5)$

These examples demonstrate how Yoruba numerals systematically expand through a multiplicative process applied to a base numeral (Item), rather than being formed by simple morpheme concatenation.

7.2 Formation Through Subtraction

The IP theory also accounts for subtractive numeration, a key feature of Yoruba numeral formation. Instead of relying solely on multiplication, Yoruba numerals often express values by subtracting smaller numbers from a larger base form. The subtractive transformation follows the pattern:

- $\text{Èédégbàáta (5,000)} \rightarrow 6,000 - 1,000$
- $\text{Èédégbègbèta (500,000)} \rightarrow 600,000 - 100,000$
- $\text{Èédégbòkèta (50,000,000)} \rightarrow 60,000,000 - 10,000,000$
- $\text{Èédégbègbàáta (5,000,000,000)} \rightarrow 6,000,000,000 - 1,000,000,000$

These numerals are formed not by simple affixation but by applying a process-based rule, making them consistent with the IP theory's model of systematic transformations rather than **Item-and-Arrangement (IA) Theory's linear morpheme stacking**.

7.3 Recursiveness and Systematic Growth

Another key feature of Yoruba numerals is their recursive nature, where the same base structure is reapplied to form increasingly larger values. This process-driven formation aligns with IP theory's transformational perspective, allowing the numeral system to scale up indefinitely.

For instance:

- $\text{Ègbèrún (1,000)} \rightarrow \text{Ègbègbèrún (1,000,000)} \rightarrow \text{Ègbègbègbèrún (1,000,000,000)}$
- $\text{Ègbègbàá (Two Million)} \rightarrow \text{Ègbègbègbà (200,000,000)} \rightarrow \text{Ègbègbègbàá (2,000,000,000)}$

This pattern continues indefinitely, illustrating a consistent and rule-governed expansion process rather than a random accumulation of morphemes.

8. Discussion

The application of Item-and-Process (IP) Theory to Yoruba numeral multiplicands provides a robust framework for understanding their morphological structure. Unlike earlier models that treated numerals as discrete morphemes combined in a fixed order, IP theory explains numeral formation as a dynamic process involving systematic transformations. This

perspective enables a more precise account of the underlying principles governing Yoruba numeration.

A critical insight from this study is the recursive nature of Yoruba numerals, which allows for the indefinite expansion of numerical values while maintaining internal coherence. By treating base numerals as foundational items that undergo systematic processes such as multiplication and subtraction, Yoruba numerals exhibit a high degree of flexibility. For example, the use of “GBẸ” (exponential marker) + “RÚN” (clipping of number five) as an exponentiation marker facilitates a structured approach to large numbers, with each additional iteration representing an order of magnitude increase (e.g., 1,000, 1,000,000, 1,000,000,000, etc.). This symbolic structure follows a specific pattern:

- One **GBẸ** + **RÚN** signifies three zeros, representing a thousand (1,000).
- Two **GBẸ** + **RÚN** represent six zeros, equivalent to a million (1,000,000).
- Three **GBẸ** + **RÚN** indicate nine zeros, equating to a billion (1,000,000,000).
- Four **GBẸ** + **RÚN** signify twelve zeros, denoting a trillion (1,000,000,000,000).

These distinctions are evident in examples such as:

- **Egbèrún** – One thousand (1,000)
- **Egbègbèrún** – One million (1,000,000)
- **Egbègbègbèrún** – One billion (1,000,000,000)
- **Egbègbègbègbèrún** – One trillion (1,000,000,000,000)

This pattern demonstrates how Yoruba numerals uniquely represent exponential values by marking groups of zeros, an approach that is both efficient and culturally resonant. It validates the Yoruba language’s inherent capacity to express figures of any size, from simple counts to vast quantities, such as quadrillions and quintillions, without compromising linguistic coherence or cultural relevance.

Furthermore, the interplay between additive and multiplicative strategies in Yoruba numeration demonstrates a complex yet intuitive system that mirrors broader cognitive patterns in numerical cognition. The ability to express values through systematic subtraction (e.g., 50,000,000 as 60,000,000 minus 10,000,000) underscores the efficiency of the system. This principle aligns with findings in morphological theory that suggest languages optimize numerical expressions for both economy and clarity.

By leveraging IP theory, this study refines previous analyses of Yoruba numeral multiplicands by demonstrating that their formation is not merely the result of compounding but a structured sequence of morphological processes. This approach not only enhances our understanding of Yoruba numeration but also contributes to broader linguistic discussions on numerical morphology, cognitive representation, and the interplay between language and mathematics.

9. Conclusion

This study analyzes Yoruba numeral multiplicands through Item-and-Process (IP) Theory, revealing them as dynamic structures shaped by systematic morphological transformations. It highlights the role of multiplication, subtraction, and exponentiation markers like *GBẸ* + *RÚN* in enabling infinite numerical expansion, demonstrating both cognitive efficiency and cultural significance. By bridging theoretical morphology with practical numeration strategies, the research underscores the value of indigenous numeral systems in linguistic studies. Additionally, it explores applications in computational linguistics, suggesting that Yoruba numeral structures can inform NLP advancements such as machine translation and automated text generation. Future research may further investigate cross-linguistic comparisons and computational modeling of indigenous numeral systems while advancing computational methods for analyzing and processing indigenous numeral systems.

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